AI, Mathematics, Education, and a public goods problem

Terence Tao · 29 October 2026

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[This is a guest post by Gireeja Ranade. This blog post was initially written in a different file format and converted using AI. — T.]

The day after the Navier-Stokes result was announced, a student came to my office hours. They said, "Can we please talk about the Navier-Stokes result and what it means? I really need some therapy." Two days ago another student mentioned, "I used to be really motivated, but now that all the models can do everything with just a prompt, I don't feel like putting in any effort at all."

Every time I have such a conversation with students, I appreciate how lucky my generation has been. We didn't have to grapple with the temptations or threats of AI when we were students. Not being able to quickly look up answers forced us to challenge ourselves, and do it together with our friends. And years of challenging ourselves somehow made us love it [1], and led us to challenging jobs across academia and industry. These jobs are simultaneously wonderful and hard. But I wonder and worry for my students, for the next generation. What do we need to do to make sure these jobs [2] are (1) available for the next generation and (2) that we have enough students who are motivated to train for them?

I've been in conversations where people are concerned about whether mathematics will survive. I'm actually not concerned about this — I'm fairly sure it will, so let us set that aside for the moment. But what I want to think about is, what do we need to do to make sure *all* mathematicians survive? And not just the `great' mathematicians, but also regular, ordinary mathematicians? Because even if we decide society only really needs the `greats [3],' we need to train many tens of thousands of students to identify who those greats are. If we want one Steph Curry, we need to make basketball exciting for a whole generation of people, and we need to support college basketball, and after-school basketball, and neighbourhood basketball.

We've seen that by treating mathematical problems as a benchmark, humans have been able to train AI models to be very good at solving hard math problems and writing proofs. Soon we will create benchmarks about `asking good questions' and `giving clear explanations' and then AI models will be good at those tasks too. They will do the thinking and the meta-thinking. This is the reality we need to plan for.

For this post I will talk about "mathematics," but I often mean mathematics broadly construed (the sciences, engineering, social sciences, and even many humanities subjects that require complex reasoning and critical thinking).

Background

People are attracted to mathematics and mathematical sciences for a variety of reasons. But importantly, whatever the motivation, mathematics has always been a `safe' subject to study. Historically, being good at math has led to a solid job and a stable lifestyle. Why? Because along the way, by studying beautiful mathematics (which in itself might not be exactly what leads to employment), the student learns critical thinking and perseverance, and becomes stronger. The transferable skills are helpful for employment.

As a result, society has encouraged students to do math, even when they might not be `great' at math. For example, as a teenager, I was part of a special program called *Gold 2002*, [4] which trained students for the International Mathematical Olympiad (IMO). I was not among the `greats' in this program. I did not make it to the IMO, unlike some of my friends. And today, I have no hope of ever winning a Fields medal. But I still enjoy doing mathematics, and (thanks to some hard work as a teenager) I'm good enough at it to be paid to do it (again broadly construed, I'm a professor in an EECS department [5]). But one of the reasons that my parents let me spend months and years working on crazy math problems was that math was a `good' thing. And so it was ok to do this, even if I didn't win any gold medals.

But today people wonder — does mathematics still have the same status? Do we need critical thinking if a prompt can do it for us? My students ask — is it worth their time to get a PhD, in mathematics, or any theoretical field, without an immediate application? Or should they focus on learning to train better deep learning models? When previously I would have many undergraduate researchers from my lab applying for PhD programs, many now prefer to go straight to industry. What do we say to them?

I have heard people say — in this new era, math will become more like the humanities. We will continue to pursue mathematics because it is beautiful and because we love it. While this may be the case, will as many students pursue it if it does not lead to a good job and a steady life [6]?

Which leads to the thought — well, does society really need all these mathematicians? We will have only our strongest, most passionate students do the very hard math, or those who love mathematics enough to do it even as `starving artists'. With AI, if one mathematician can do the job of five or ten, should we shrink our departments and our graduate programs? What jobs do we expect our students to get when they graduate? And if there are no jobs for `thinkers,' and `intellectual explorers', then why should students pursue a path towards a PhD in math or a theoretical science?

This brings us to a public goods problem, which I elaborate on further below. I believe it is essential for us, the human society, to maintain our intellectual culture. And for this we need lots of students getting degrees in mathematics, sciences, and the humanities, but we need to find ways to actually motivate the students to want these degrees and the associated training.

What is our culture and why is it important?

As a child, I would see a big splash in the newspaper about the Nobel prizes every year. I didn't understand the science or why the literature was meaningful. I just understood that this stuff was important, and that humans valued creativity, expression and intellectual and scientific achievement. And until recently, I had never consciously wondered about the cultural importance of Alfred Nobel establishing those prizes. I had always taken for granted that we

should strive for excellence… but I see now that this was an impact of our culture, and that there are so many conscious decisions that have led to this culture.

But now, people have begun to worry about `The end of the age of heroes'. In his blog post, Noah Smith makes multiple interesting points, but I will paraphrase one: even as mathematicians will likely retain their jobs as math-understanders [7] (since mathematicians are relatively cheap), they will lose their status as heroes.

My question is: while it may be ok for the mathematician to lose their hero status, what impact does this have on the next generation? In my younger days, and even now, I know I benefitted from being around, and hearing about, heroes — the brilliant, blazingly quick, hardworking minds, with the humblest of personalities to match. It gave me someone to look up to and goals to aspire to. With heroes gone, how would we orient our cultural compass?

Aspects of our culture are built on respect for facts about the universe. Nvidia CEO Jensen Huang said in an interview with Ezra Klein a few weeks ago that it doesn't matter if the multiplication table is starting to be forgotten, and that "we're going to lose some finer intellectual dexterity … I don't know how valuable it is for most people to learn how to do surface integrals or partial differential equations. I don't really know how important that is, but it's important to some people."

I was bothered reading this, because I was worried about the implications for our culture.

Multiplication tables matter not just for their own sake but for what they represent [8]. Yes, knowing 11×12 builds fine intellectual dexterity. But they also show young children that there are fundamental, unchanging facts about the world. Invariants. And starting from this, we can build up confidence that allows kids to grow into students who are grounded and questioning because they can both see and understand why there are invariants [9].

Our understanding of partial differential equations allowed us to develop numerical computer programs that can solve them for us. And when the computer does solve them, it is amazing that we (as a human society, not every individual) are able to understand exactly what the computer did. But this is not true when AI does all the work for us; we lose the essential human understanding. I will come back to the value of this soon.

It is important to maintain a culture where we value and respect intellectual dexterity and expertise. And admire the young children who know their multiplication tables, and the older people who win Fields medals and Nobel prizes. And respect those people who care deeply about partial differential equations, and create space for them to develop their expertise, even if not everyone understands them. The goal of expanding and developing human expertise in subjects can and should be independent of the state of AI [10]. So that we can ensure generations of students will continue to want to study and experience the "unreasonable effectiveness" of mathematics.

Education: Canary in the coal mine

Every professor and teacher I know has grappled with AI-based academic dishonesty by students. Students cheat when `doing it the right way' doesn't seem worth the time and effort. And because our culture for students (especially in the United States) has focused on achievements/badges (or, as I tell my students, `sticker collection'), instead of building true

understanding and intellectual depth, students want to do more and do it faster, even if they are not doing it deeper or better.

We need to realize that this is just the canary in the coal mine. Students are turning in proofs/essays/code they don't fully understand. But this behavior is not limited to academic settings. It is likely that engineers are checking in code they didn't fully read. It is only a matter of time before a researcher publishes a result they don't fully understand, or an administrator signs off on a decision they don't fully comprehend. And soon enough, AI systems will be making ever more complex decisions that we just sign off on, because we don't understand the foundation of the system and we gave up questioning.

If we go down the slippery slope, we risk becoming AI-sock puppets [11]. Faced with AI-output, we will take quick looks and "sanity checks" to make sure AI output is reasonable and move on, but it won't have the same gravitas and deliberation that comes from a human-authored proof/document. We will have no choice but to use AI, because that will be the only way to keep up. Sanity checks themselves may become illusions — because we will be out of practice at true understanding. It is likely that we will have verified Lean proofs that assure us of correctness, but now we have to evaluate the faithfulness of the theorem statements. (And Lean itself has a very steep learning curve. There is a lot of work to be done to make sure such verification tools are universally accessible to all [12].)

The only thing that can protect us against this is culture. A culture where the point of pride is in the understanding of the work and in the exposition of it. A culture where it is ok to say we don't understand, ask for clarification, or (respectfully) push back. And where such requests are valued (instead of saying, "let us move on, we only have a few minutes").

A culture that appreciates being slow and steady and careful. A culture where a student can pursue a more theoretical research area for their PhD, but also be assured that this does not destroy their chance to get a stable, enjoyable job.

So how do we cultivate this culture?

Such culture building must start in the classroom. We must teach harder, better, and stronger [13]. We need to train students so that they know what they know and don't know — students who can identify proof by intimidation and by authority, and have the confidence to push back. We should provide homework problems that let students explore concepts in depth, and force them to confront ideas they do not understand. Students thrive when they are challenged (with support) and then succeed. This needs resources.

We must walk away from classes that teach students the "high-level ideas" but do not expect students to fully grasp the rigorous details "because AI can do this stuff nowadays, [14]" or because "it is too hard for the students [15]." Because these can create a generation of students who don't know the difference between understanding things deeply versus superficially. And then this can propagate.

This is now a bigger deal than it was a few years ago. It used to be that knowing a solution strategy or the right algorithm to apply was sufficient for students to become productive members of society. But now that this work can be done by AI systems, we have to be able to understand what the AI has generated, process it and question it, which requires not just

knowing the names of the algorithms but also the proofs of why they work, what the key assumptions are, and where and how they break down.

A theorem may be correct given its assumptions. But the gap between assumptions and what we actually care about might admit counterexamples. Identifying such counterexamples requires knowing the modeling and the math at multiple levels. Interpretation matters and ambiguity can creep in when identifying if a formal Lean statement is faithful to the natural language. For such tasks mathematical mental muscles are essential.

Amit Sahai, Po-Shen Loh, and others have beautifully argued why we are going to need more (highly trained) mathematicians. I won't repeat their argument here. But I will add to their point — it is important that we have a large, diverse set of people, with different life paths and trajectories, becoming mathematicians. People who do not all think exactly the same way, who can interpret, debate, disagree and convince one another.

A public goods problem

This brings us to a public goods problem. We need to maintain our culture. For this, we need to engage and excite lots of different students. We know that if we don't incentivize students to learn, many of them won't, and our (humankind's) mental muscles will atrophy. We know that when humans moved from working in the fields to having indoor desk jobs, our bodies suffered. And to compensate, we invented things like the gym to make sure our bodies stayed fit.

But there is a catch. We pay to go to the gym, not the other way around. Entry-level jobs in fact used to train people while also getting useful contributions from them. Entry-level jobs, or PhD programs, are like being paid to go to the gym. If these are taken away by AI, how do we train people? And how do we convince people that they should be trained?

There is a tremendous investment needed in thinking about what future sustainable solutions to this will look like. This is just the beginning of that conversation. But in the short term, it is important culturally to celebrate and reward the work of understanding and communicating mathematical results — and not just by mathematicians but by society at large. We also need to advocate for educational institutions at all levels and for what they represent for our culture. Teachers need more resources, not less, to be able to teach in the AI-age, because we need to be teaching more, not less. There is no silver bullet. AI tools alone cannot solve problems in the education sector — these come with their own set of challenges. We should be able to offer individualized, 1-1 human tutoring to help students at different levels. I hope we can collectively build a positive future, where we use AI-tools effectively to help us in ways that raise everyone up together.

As a concrete first step in this direction, I invite university leaders to declare their commitment to hiring across departments in the sciences, engineering, social sciences and humanities, and announce their unwavering support for human intellectual growth and exploration. Similarly I invite philanthropists and funding agencies to declare their support for open-ended research, theoretical pursuits and rigorous, grounded education in the fundamentals. Students need to know that we have got their back, and that we are working for their future — because they are our future.

I would like to thank Terry Tao and Nestor Guillen for the invitation to write this post, and also Ted Theodosopoulos, whose 1st Relatorium dialogue on Mathematics and AI started part of

this conversation. Thanks also to Anant Sahai for many conversations that have helped refine these thoughts, as well as Vinayak Ranade, Sukhada Fadnavis, Shubhangi Saraf and Amit Sahai for their helpful feedback. Thank you also to the organizers and participants of the Renaissance Philanthropy AI for Math Workshop 2026 — conversations there also influenced my thoughts.

Notes

[1] I understand that challenge and struggle does not work for all students, but it does help a good number. In this post, I will focus on that set. There is a longer discussion about education to be had more broadly at another time.

[2] Jobs that are typically held by PhDs that apply mathematical/other critical thinking skills. I understand that there are far more complex questions regarding jobs in the age of AI, and I'm not trying to tackle them. For now I focus on the narrow question of jobs for highly-trained scientists/mathematicians/humanists.

[3] I argue that we need far more than just the `greats,' but just hypothetically.

[4] I am forever indebted to M. Prakash sir, out of the Bhaskaracharya Pratishthan. He intensively trained 200+ highschoolers across Pune, India, and several from the group went on to win IMO medals (another example of the Steph Curry argument). And the rest of us benefitted on the side. It was a privilege to participate.

[5] I teach courses on the mathematical side of the department, such as optimization, probability, and deep learning.

[6] I also love to dance, it is beautiful. But I am nowhere good enough to earn a living doing this. Luckily the math worked out ;-).

[7] Human understanding can never be developed by an AI, no matter how good AIs get. AI-augmentation of humans cannot generate human understanding.

[8] Perhaps we will find skills that are a suitable substitute for multiplication tables, that do everything that tables do and are `better' in other ways. I'm ok with teaching those skills instead.

[9] As Picard said "there are four lights."

[10] And every PhD granted is a new human expert in a subject.

[11] Not quite a meat proxy, but still bad.

[12] Some of our work to make lean accessible is available at leantutor.org

[13] not faster. https://www.youtube.com/watch?v=gAjR4_CbPpQ

[14] There is a space in this system for courses that teach students how to "use" tools (e.g. prompt engineering courses that have recently come up at many community colleges in California), or general AI literacy courses. But we must not confuse the role of these courses. Those may be helpful (in the current climate) for maintaining employment. But they do not replace the role of courses that fundamentally teach critical thinking, and should not take resources away from such courses.

[15] To cater to the fact that students come in at different levels, we need to have scaffolding classes at the lower-division level. A student may take longer to graduate, but should be able to take even the hardest class given enough time and hard work.

Comments (10)

 **frobitz13** · 30 October 2026

Thank you for this essay, it expresses thoughts I have had more eloquently than I could.

 · 30 October 2026

In my humble opinion, the first thing to do is to understand the capabilities and limits of AI models, more specifically LLMs. Assuming that they can do anything that mathematicians can do would be severely wrong. Let's recall what **Sun Tzu** once said: *"If you know the enemy and know yourself, you need not fear the result of a hundred battles. If you know yourself but not the enemy, for every victory gained you will also suffer a defeat. If you know neither the enemy nor yourself, you will succumb in every battle."*

>  **old-bielefelder** · 30 October 2026
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> It is impossible to keep the LLMs fixed. They are like running boars, as hunters would call them. For instance, the steps from ChatGPT 5.2 to 5.4, 5.5 and then 5.6 all were tremendous. Each time I had to "reinvent" my prompting. But in some sense that also was part of the fun.

 · 30 October 2026

I would hold off on making assumptions on whether AI can ask the right questions. This may well happen, but it is not clear when or whether the system will be able to ask the right questions and develop new areas on its own without human oversight. If so, it is game over most likely for all of us, but I would be careful in assuming this will be the case. More likely mathematicians will be needed to steer matters in the right direction. This requires deep domain knowledge combined with creative exploration, and possibly a shift in focus from developing an existing domain, i.e. proving results in the same direction, to a different type of exploration, moving beyond or across the boundaries of what usually determines an expert. This combination of creativity and analytical prowess can form the focus of future education, where thus far most of the focus is on developing theorem proving muscle. I agree that our mathematical culture is based on mentoring and apprenticeship on a large scale. I don't see this alter due to the argument put forward in the article: it takes many players to build a new generation of excellent mathematicians.

>  · 30 October 2026
>
> "I would hold off on making assumptions on whether AI can ask the right questions. This may well happen, but it is not clear when or whether the system will be able to ask the right questions **and develop new areas on its own without human oversight**."
>
> It's no longer an assumption.
>
> Commenter: "Scott [Aaronson], can you give a number 1-10 for how novel she takes the techniques used in the proof to be? Is it a clever combination of old ideas?"

> Scott: “She’s described the proof to me as consisting of a totally alien new error-correcting code — unlike any that human PCP experts had proposed — plus a lot of standard, well-understood techniques to lift that code to a PCP and hence a proof of the UGC (with the latter, more “boring” part being the part that takes most of the time to verify).” See his blog post.

 **Connelly Barnes** · 30 October 2026

Thanks for the interesting post. I feel as a long-time believer in the AI Singularity (since at least the late 1990s) and a computer scientist (undergrad math + physics major) that it’s inevitable that AI will allow elegant human-comprehensible proofs to be synthesized when they exist. Respectfully, I feel the perspective in this article is still too focused on math proofs as a timeless artifact, which is a nice and philosophical way to look at math, but I feel maybe less related to employment going forward.

In contrast, I feel if the scaling laws for FLOPS/dollar hold up due to the underlying physics allowing us to skate further along the AI scalability curves, then we will see proofs going forward as more of a context-dependent “vibe-proven” thing where you spin them up on the fly. The computer science analogy is that one can construct proofs over the integers, reals, and floats in the Z3 theorem prover, and they will be right, but no one will read them. Why would one read them? That’s not the point. The point is to understand if some axiomatic system associated with some (usually) complex language (e.g. a compiler) or system (e.g. a flight system, nuclear power plant, business plan, an exploration of outer space) is logically consistent, or safe, or to understand how the axioms and the proof connect (i.e. are all axioms necessary). So I feel in this perspective, mathematicians’ value is going to be less about the artifact (the proof) than about understanding, contextualizing, or changing the underlying assumptions and goals, as some kind of human <-> AI <-> formal system bridge.

 **Connelly Barnes** · 30 October 2026

To continue the argument (sorry, the button posted my comment before I was finished), we don’t want nuclear power plants to fail, or planes to fall out of the sky, or bridges to fall down because humans or AIs used a bunch of unrealistic assumptions to put together a complicated system that fails, despite being provably correct. So said another way, I feel that applied math and physics will still be quite relevant, but the heroes may be good communicators who apply AI systems and connect them to the physical world and physical laws in reasonable ways.

 **Enno** · 30 October 2026

Yes, humans will be in charge of realistic input and intermediary verification feedback, grounded in physial reality, to ensure useful output, results that align with user’s intentions they formed from their reality

 **Nora** · 30 October 2026

Pointing to neighbourhood basketball to explain why we need ordinary mathematicians alongside the greats struck me as a thoughtful argument. High-level breakthroughs rely on a broad ecosystem where people simply enjoy the practice. Without thousands of students exploring math for its own sake, the foundation for deeper discoveries quickly disappears.



Rakesh Kalra · 31 October 2026

The part about students turning in work they don’t fully understand feels close to what parents see with homework and AI. Kids need room to use these tools while still being able to explain the answer in their own words.